

BCM Project Task Breakdown by Team

1. Embedded & Diagnostics Team - Task List (Simulation-Based Project)

Phase 1: Setup & Requirements

- Understand BCM features (e.g., door lock, light control, wiper)
- List control signals to be implemented
- Study UDS protocol (0x10, 0x22, 0x19)
- Understand CAN protocol and message structure
- Decide on microcontroller (e.g., STM32F4)

Phase 2: Project Setup

- Configure STM32CubeMX (CAN, UART, GPIO)
- Generate and import code to STM32CubeIDE
- Organize folders for CAN, UDS, DTCs

Phase 3: CAN Protocol Integration

- Initialize and test CAN communication
- Create message IDs for BCM functions
- Implement send/receive routines

Phase 4: UDS Implementation

- Implement 0x10 (Diagnostic Session Control)
- Implement 0x22 (Read Data By ID)
- Implement 0x2E (Write Data By ID)
- Optional: Implement 0x19 (Read DTCs)

Phase 5: Diagnostics & DTCs

- Define faults (e.g., sensor failure)
- Assign DTC codes
- Simulate fault detection and DTC storage
- Implement UDS 0x19 response

Phase 6: Testing & Debugging

- Test each service using simulated tools
- Validate system behavior
- Support Software team in integration

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Phase 7: Final Integration

- Share simulation results and documentation
- Prepare demo/testing script

2. Software & Application Team - Task List

Phase 1: Setup

- Understand UDS services & CAN flow
- Decide testing method (Python script, GUI, etc.)

Phase 2: Simulation/Interface

- Build Python tool to simulate tester
- Optional: Build GUI to interact with Embedded code
- Add fields for service ID input, DTC view, and command sending

Phase 3: Testing

- Create and run test cases for UDS services
- Trigger faults and verify DTC logging
- Share testing data and results

Phase 4: Support & Handover

- Collaborate with Embedded team for debugging
- Share tools/scripts with Documentation team
- Provide screenshots and logs for reports

3. Planning & Documentation Team - Task List

Phase 1: Planning & Setup

- Create Kanban board (JIRA/Excel)
- Add all tasks by team and assign owners
- Track progress weekly

Phase 2: Documentation

- Write system overview and team roles

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- Maintain UDS services, CAN IDs, and DTCs log
- Document test plans and results

Phase 3: Reporting & Review

- Prepare presentation slides
- Create architecture flowcharts & communication diagrams
- Record demo walkthrough if needed